Dr Maria Zamyatina — CV

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Academic career

Postdoctoral Research Fellow — Aug 2025-now
Department of Physics and Astronomy, University of Exeter | Exeter, UK
Maternity leave (9 months)

Postdoctoral Research Fellow — Apr 2022-Jul 2025
Department of Physics and Astronomy, University of Exeter | Exeter, UK

Postdoctoral Research Fellow — Sep 2019-Mar 2022
Department of Physics and Astronomy, University of Exeter | Exeter, UK

Education

PhD in Environmental Sciences — 2015-2020
School of Environmental Sciences, University of East Anglia | Norwich, UK
Supervisors: Prof. Claire Reeves, Dr Paul Griffiths, Dr Marcus Köhler, Dr Mike Newland
Thesis: Impacts of C_1-C_3 alkyl nitrates on tropospheric ozone chemistry

MSc in Climate Change with Distinction — 2014-2015
School of Environmental Sciences, University of East Anglia | Norwich, UK
Supervisor: Prof. Claire Reeves
Thesis: Investigation of the relationship between tropospheric ozone production efficiency and carbon bond emissions

Specialist Diploma in Meteorology — 2009-2014
Faculty of Geography, Lomonosov Moscow State University | Moscow, Russia
Supervisor: Prof. Alexander V. Kislov
Thesis: Climatically-induced variations of the Caspian Sea level over the last Millennium

Publications

15. Ahrer, E-M., Gandhi, S., Alderson, L., Kirk, J. et al. (incl. **Zamyatina, M.**) (2025). Tracing the formation and migration history: molecular signatures in the atmosphere of misaligned hot Jupiter WASP-94Ab using JWST NIRSpec/G395H. MNRAS.
14. Meech, A., Claringbold, A. B., Ahrer, E-M., Kirk, J. et al. (incl. **Zamyatina, M.**) (2025). BOWIE-ALIGN: Sub-stellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy. MNRAS.
13. Kirk, J., Ahrer, E-M., Claringbold, A. B., **Zamyatina, M.**, Fisher C. et al. (2025). BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b. MNRAS.
12. Kirk, J., Ahrer, E-M., Penzlin, A. B. T., Owen, J. E. et al. (incl. **Zamyatina, M.**) (2024). BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history. RAS Techniques and Instruments.
11. Penzlin, A. B. T., Booth, R. A., Kirk, J., Owen, J. E. et al. (incl. **Zamyatina, M.**) (2024). BOWIE-ALIGN: how formation and migration histories of giant planets impact atmospheric compositions. MNRAS.
10. Espinoza, N., Steinrueck, M., Kirk, J., MacDonald, R. J. et al. (incl. **Zamyatina, M.**) (2024). Inhomogeneous terminators on the exoplanet WASP-39 b. Nature.
9. Christie, D. A., Mayne, N. J., **Zamyatina, M.**, Baskett, H., Evans-Soma, T. M., et al. (2024). Longitudinal filtering, sponge layers, and equatorial jet formation in a general circulation model of gaseous exoplanets. MNRAS.
8. **Zamyatina, M.**, Christie, D. A., Hébrard, E., Mayne, N. J., Radica, M. et al. (2024). Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example. MNRAS.
7. Taylor, J., Radica, M., Welbanks, L., MacDonald, R. J. et al. (incl. **Zamyatina, M.**) (2023). Awesome SOSS: atmospheric characterisation of WASP-96b using the JWST early release observations. MNRAS.
6. Radica, M., Welbanks, L., Espinoza, N., Taylor, J. et al. (incl. **Zamyatina, M.**) (2023). Awesome SOSS: transmission spectroscopy of WASP-96b with NIRISS/SOSS. MNRAS.
5. **Zamyatina, M.**, Hébrard, E., Drummond, B., Mayne, N. J., Manners, J. et al. (2023). Observability of signatures of transport-induced chemistry in clear atmospheres of hot gas giant exoplanets. MNRAS.
4. Ridgway, R. J., **Zamyatina, M.**, Mayne, N. J., Manners, J., Lambert, F. H. et al. (2023). 3D modelling of the impact of stellar activity on tidally locked terrestrial exoplanets: atmospheric composition and habitability. MNRAS.

3. Christie, D. A., Lee, E. K. H., Innes, H., Noti, P. A. et al. (incl. **Zamyatina, M.**) (2022). CAMEMBERT: A Mini-Neptunes GCM Intercomparison, Protocol Version 1.0. A CUISINES Model Intercomparison Project. Planet. Sci. J.
2. Braam, M., Palmer, P. I., Decin, L., Ridgway, R. J., **Zamyatina, M.** et al. (2022). Lightning-induced chemistry on tidally-locked Earth-like exoplanets. MNRAS.
1. Gromov, S.A., Gromov, S.S., **Zamyatina, M.**, Trifonova-Yakovleva, A. M. (2013). First-order evaluation of transboundary pollution fluxes in areas of EANET stations in Eastern Siberia and the Russian Far East. EANET Science Bulletin, 3:195-203.

Invited talks

Jul 2025	Atmospheric chemistry in the Solar System and beyond (review talk) Exoclimes VII conference | Montreal, Canada
Mar 2024	Overview of the Met Office Unified Model configuration for hot Jupiter atmospheres International Space Science Institute (ISSI) workshop | Bern, Switzerland
Feb 2024	Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere University of Bristol (astronomy seminar) | Bristol, UK
Feb 2023	Atmospheric dynamics and chemistry on exoplanets University of Queensland (astronomy seminar) | Brisbane, Australia University of Southern Queensland (exoplanet seminar) | Brisbane, Australia University of New South Wales (astronomy seminar) | Sydney, Australia
Nov 2022	Observability of signatures of wind-driven chemistry in atmospheres of hot gas giants Ludwig Maximilian University (exoplanet group seminar) | Munich, Germany Celebrating JWST's first six months of exoplanet data workshop | Ringberg castle, Germany
Oct 2022	Modelling chemistry of hot Jupiter atmospheres with the Met Office Unified Model Met Office | Exeter, UK
Feb 2022	Transport-induced quenching shapes transmission spectra of warm and hot Jupiters University of Warwick (astronomy seminar) | virtual

Contributed talks

Sep 2023	Metallicity masquerade: how to use quenching to distinguish between different planet metallicities University of Bristol (BOWIE meeting) | Bristol, UK
June 2021	Overview of the Met Office Unified Model adapted to simulate exoplanetary atmospheres Ariel consortium meeting | virtual
Apr, Sep 2021	3D simulations of warm and hot Jupiter atmospheres: the role of 3D mixing in shaping CH_4-to-CO conversion pathways EPSC conference | virtual UKEXOM conference | virtual University of Exeter (astronomy seminar) | Exeter, UK
Mar, Apr, Jun 2019	Impact of C_1-C_3 alkyl nitrate chemistry on tropospheric ozone: box and global model perspectives University of Exeter (XCS seminar) | Exeter, UK EGU conference | Vienna, Austria University of East Anglia (AMB seminar) | Norwich, UK
Apr 2017	Adding new chemistry into UM-UKCA Cambridge-EnvEast doctoral alliance symposium | Cambridge, UK
Sep 2012	Assessment of climatological potential of transboundary air pollution transport in Eastern Siberia and the Russian Far East Air quality management at urban, regional and global scales 4th international symposium/IUAPPA regional conference | Istanbul, Turkey

Posters

Jul 2025	Spatial variability in CH_4-CO interconversion pathways in hot Jupiter atmospheres Exoclimes VII conference | Montreal, Canada
Apr, Jun 2024	Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere UKEXOM conference | Birmingham, UK Exoplanets 5 conference | Leiden, Netherlands
Sep 2022	Applying known chemical kinetics data to model atmospheres of extrasolar planets iCACGP-IGAC conference | Manchester, UK
Sep 2021	Local and global impacts of C_1-C_3 alkyl nitrate photochemistry and emissions on tropospheric ozone IGAC conference | virtual
Sep 2018	Impact of alkyl nitrate chemistry on tropospheric ozone iCACGP-IGAC conference | Takamatsu, Japan
Mar, Apr 2018	Impact of C_1-C_5 alkyl nitrate chemistry on tropospheric ozone - a box modelling study Cambridge-EnvEast doctoral alliance symposium | Cambridge, UK EGU conference | Vienna, Austria

Awards

2023	Above & Beyond Award	
2022	**EPSRC vacation internship** (for 3 interns)	12893.55£
2022	**Jackson-Grime-Davies (JGD) research internship** (for 1 intern)	2428.71£
2021	**IGAC Early Career Scientist poster prize & travel grant**	1227.70£
2015-2019	**Lord Zuckerman studentship**	112269.50£
2014-2015	**Simon Wharmby postgraduate scholarship**	3000.00£
2012	**World Meteorological Organization travel grant**	1154.10£

Awarded observing time

Feb 2024 JWST's exoplanet grand tour spectroscopic survey
- Co-I HST GO-17612 (PI: David Sing) — 24 orbits
- Co-I JWST GO-5924 (PI: David Sing) — 125.70 hours

Feb 2024 Starspots, hazes, and disequilibrium chemistry: a deep dive into the atmosphere of HAT-P-18b
- Co-I JWST GO-5844 (PI: Michael Radica) — 16.40 hours

May 2023 Putting it all together: Dynamics and chemistry probed through transmission spectroscopy of a cloud-free exoplanet
- Co-I JWST GO-4082 (PI: Michael Radica, Co-PI: Jake Taylor) — 6.69 hours

May 2023 Hot Jupiter atmospheric forecast: are mornings cloudier than evenings in other worlds?
- Co-I JWST GO-3969 (PI: Nestor Espinoza, Co-PI: Diana Powell) — 61.53 hours

May 2023 Does atmospheric composition actually trace formation? Observing aligned vs misaligned hot Jupiters as a testbed
- Co-I JWST GO-3838 (PI: James Kirk Co-PI: Eva-Maria Ahrer) — 49.21 hours

May 2023 Testing the C/O ratio prediction for hot Jupiters from disk-free migration
- Co-I JWST GO-3154 (PI: Eva-Maria Ahrer) — 10.36 hours

Supervision

Primary supervisor and co-supervisor. Students who went on to do a PhD are marked with *.

PhD supervision (2)

Sep 2024-now Harry Baskett
Thesis: TBD
Co-supervisors: Dr. E. Hebrard, Prof. N. J. Mayne

Nov 2020-May 2023 Robert J. Ridgway
Thesis: Simulating the impact of stellar flares on the climate and habitability of terrestrial Earth–like exoplanets
Co-supervisors: Prof. N. J. Mayne, Prof. F. H. Lambert, Dr. J. Manners

Undergraduate and summer internship supervision (4)

Jun-Aug 2022 EPSRC-funded: Harry Baskett*, Ben Moore*, James McDermott*; JGD-funded: Graig Lils
Project: 3D modelling of hot Saturn atmospheric chemistry

Teaching

Jul 2023 **Module leader**
Module: No place like home: placing Earth in its geological and astronomical contexts
International sustainability summer school | University of Exeter, Exeter, UK

Jul 2022, Jul 2023 **Lecturer**
Module: No place like home: placing Earth in its geological and astronomical contexts
International sustainability summer school | University of Exeter, Exeter, UK

Sep 2021-Feb 2022 **Associate Tutor**
Modules: Experimental science, Frontiers in science
University of Exeter | Exeter, UK

Jan 2018 **Instructor**
Module: Introduction to Python in Environmental Sciences
University of East Anglia | Norwich, UK

2015-2018 **Associate Tutor**
Modules: Numerical skills for scientists, Atmospheric chemistry and global change, Atmospheric composition (measurements and modelling), Atmosphere & oceans I
University of East Anglia | Norwich, UK

Academic community

Organisation of scientific meetings

26-30 Jun 2023 Exoclimes VI conference (LOC member, session chair) — ~200 attendees
University of Exeter | Exeter, UK

22-24 Jun 2023 ExoSLAM school (LOC member) — ~50 attendees
University of Exeter | Exeter, UK

5-6 Dec 2022 BOWIE meeting (co-organiser) — 17 attendees
University of Exeter | Exeter, UK

Sep 2017-Jun 2018 Atmospheric and Marine Biogeochemistry (AMB) seminars (co-organiser) — ~20 attendees

University of East Anglia | Norwich, UK

Reviewing

Journals: The Astrophysical Journal, Monthly Notices of the Royal Astronomical Society

Proposals: JWST Cycle 3 TAC external expert, JWST Cycle 4 TAC panelist

Outreach

Jul 2024 Joint press release about Espinoza et al. (2024) paper:
Research confirms that distant world's eternal sunrise and sunset are not alike

Sep 2023 Expert scientist at the Climate Exhibition (part of the British Science Festival)

Nov 2015-Jun 2019 Maintainer of @AtmosChemUEA Twitter account

Vocational Experience

Aug-Sep 2013 **Weather Forecaster**
Sheremetyevo International Airport | Moscow, Russia

Jun-Jul 2013 **Technician**
Department of Actinometry, Meteorological Observatory
Lomonosov Moscow State University | Moscow, Russia

Vocational training

Sep 2023 Belbin training
Mar 2023 Leadership training
Dec 2022 Interview training
Sep 2021 Learning and Teaching in Higher Education (LTHE) Unit 1
Mar 2020 JWST proposal planning workshop
Apr 2016 NAME workshop
Jan 2016 Introduction to UKCA
Dec 2015 Introduction to Unified Model
Nov 2015 Introduction to Atmospheric Science
2015-2019 EnvEast Doctoral Training Programme